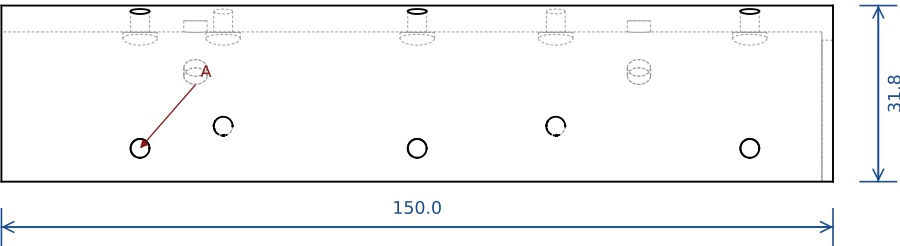
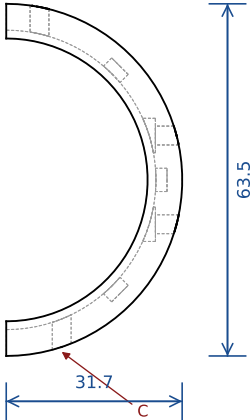


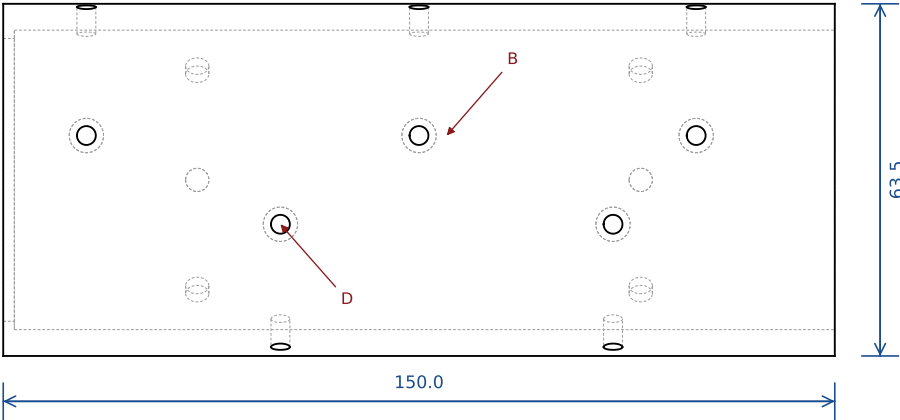
TOP



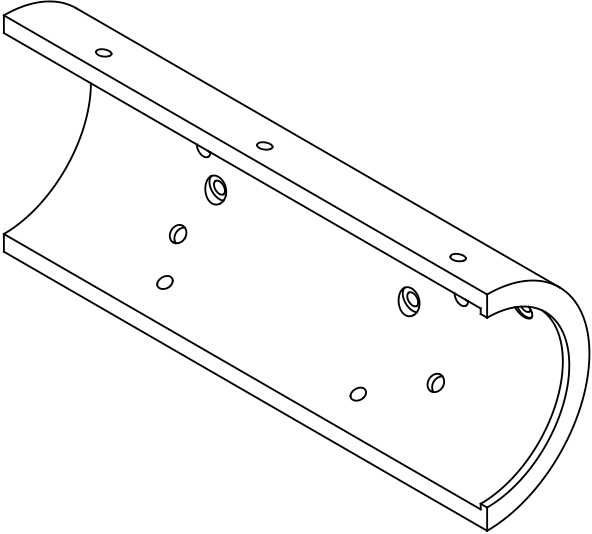
RIGHT



BACK



ISO (not to scale)



NOTES

- Pair of halves from one saw-cut tube, split on the y=0 plane, line-bored together to Ø54.0
- All screw counterbores are cut from the BORE side (heads sit under the liner)
- Rear liner lip: x 148..150, inward to R25.5
- Top rows feed A1 (x 25/70/130); bottom rows feed A3 (x 40/100). Do not mix them up

HOLE / FEATURE SCHEDULE

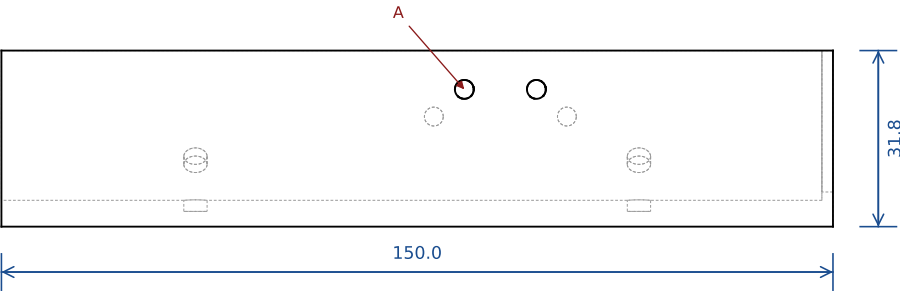
- A: 3x Ø3.4 THRU vertical at y=+6 (top crown row), Ø6.2 C'BORE 2.3 deep from the bore @ x 25, 75, 135
- B: 3x Ø3.4 THRU horizontal (+y) at z=+8 (top skirt row), Ø6.2 C'BORE from the bore @ x 25, 75, 135
- C: 2x Ø3.4 THRU vertical at y=+10 (bottom crown row), Ø6.2 C'BORE from the bore @ x 40, 100
- D: 2x Ø3.4 THRU horizontal (+y) at z=-8 (bottom skirt row), Ø6.2 C'BORE from the bore @ x 40, 100

GENERAL

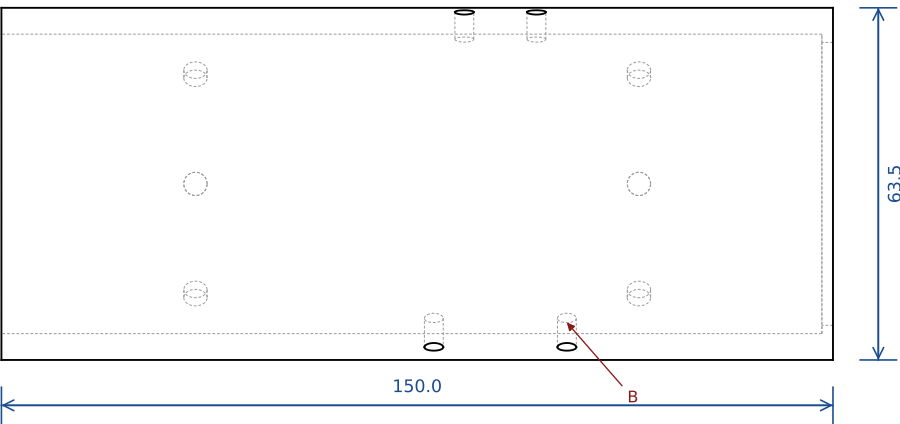
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.  
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.  
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

|                                                                                            |                                                                  |                                                                                            |                                              |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>RFID BIKE LOCK - CNC CASING</b><br>cad/cnc_casing_cq.py rev 3b                          | MATERIAL<br>6061-T6 tube 2.5" OD x 3/16" wall, or PETG (stage 1) | SCALE (ortho views)<br>0.73 : 1                                                            | DATE<br>2026-09-07                           |
| <b>C1 - chassis half (carries the boxes)</b><br>file: cnc-design/step/C1_chassis_half.step | QTY<br>1                                                         | BOUNDING BOX x / y / z<br>150.0 / 31.8 / 63.5 (x -0.0..150.0, y -0.0..31.8, z -31.8..31.8) | SHEET<br>1 / 12<br>3rd angle<br>A3 landscape |

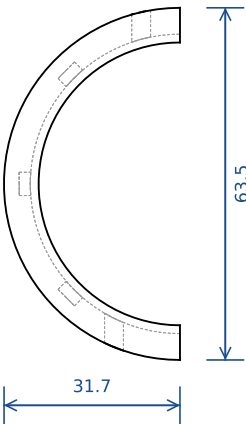
TOP



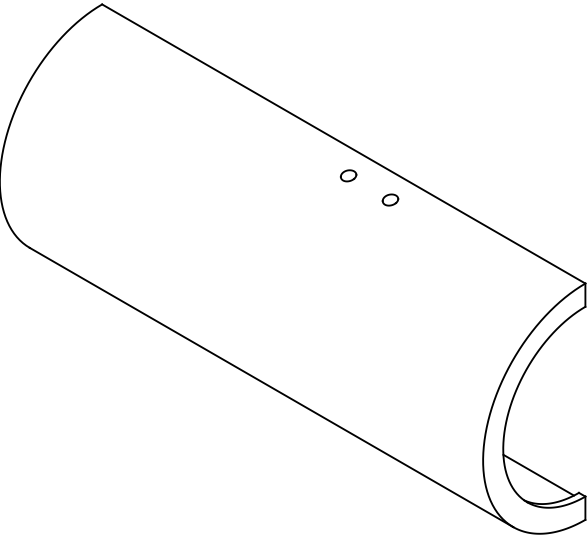
FRONT



RIGHT



ISO (not to scale)



NOTES

- Mate of C1 (same tube, same line-bore)
- Counterbores from the BORE side
- Rear liner lip: x 148.,150, inward to R25.5

HOLE / FEATURE SCHEDULE

A: 2x Ø3.4 THRU vertical at y=-7 (closure block), Ø6.2 C'BORE from the bore @ x 83.5, 96.5

B: 2x Ø3.4 THRU vertical at y=-11.9 through the BOTTOM wall (hinge block), Ø6.2 C'BORE from the bore @ x 78, 102

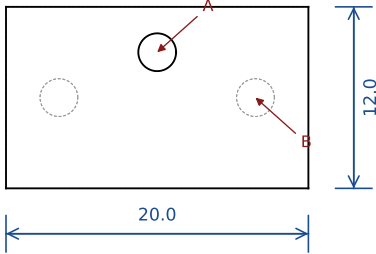
S: 6x Ø4.2 +0.1/-0 x 2 BLIND radial, from the bore (liner press-fit studs) @ x 35, 115 at -45/0/+45 deg from the mid-arc (6 per half)

GENERAL

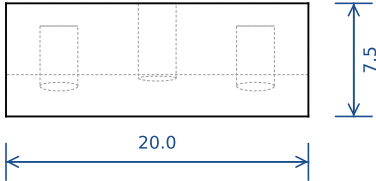
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.  
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.  
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

|                                                                             |                                                                  |                                                                                            |                                              |
|-----------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>RFID BIKE LOCK - CNC CASING</b><br>cad/cnc_casing_cq.py rev 3b           | MATERIAL<br>6061-T6 tube 2.5" OD x 3/16" wall, or PETG (stage 1) | SCALE (ortho views)<br>0.73 : 1                                                            | DATE<br>2026-09-07                           |
| <b>C2 - clamp half (hinged)</b><br>file: cnc-design/step/C2_clamp_half.step | QTY<br>1                                                         | BOUNDING BOX x / y / z<br>150.0 / 31.8 / 63.5 (x -0.0..150.0, y -31.8..0.0, z -31.8..31.8) | SHEET<br>2 / 12<br>3rd angle<br>A3 landscape |

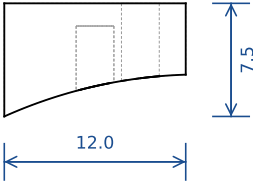
TOP



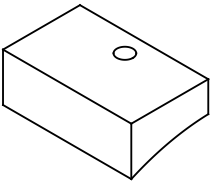
FRONT



RIGHT



ISO (not to scale)



NOTES

- Underside is a saddle R31.8 - sits on C2's OD
- Top face is flat and CONTACTS the A1 pocket roof under the closure screw
- The closure screw comes down the latch bore: M3x8 low-head

HOLE / FEATURE SCHEDULE

A: M3 closure-screw thread THRU, M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG) @ (90, -4)

B: 2x M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG), from the saddle side, 6 deep @ x 83.5, 96.5 at y=-7

GENERAL

Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.  
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.  
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

RFID BIKE LOCK - CNC CASING

cad/cnc\_casing\_cq.py rev 3b

MATERIAL  
steel or 6061 (PETG stage 1)

SCALE (ortho views)  
2.00 : 1

DATE  
2026-09-07

B - closure block (on C2, under A1)

file: cnc-design/step/closure\_block.step

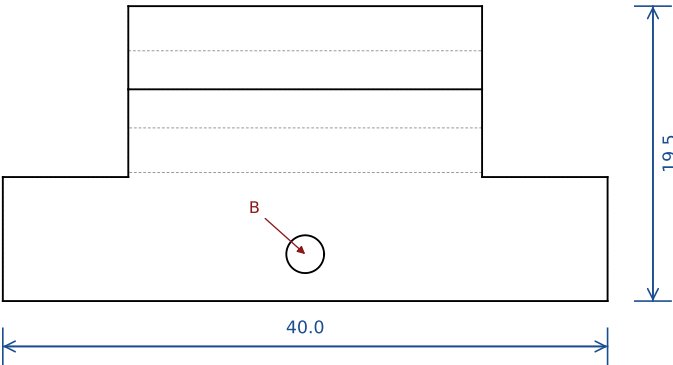
QTY  
1

BOUNDING BOX x / y / z  
20.0 / 12.0 / 7.5 (x 80.0..100.0, y -13.0..-1.0, z 29.0..36.5)

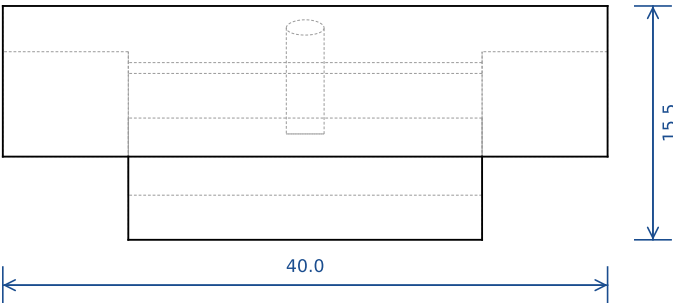
SHEET  
3 / 12

3rd angle  
A3 landscape

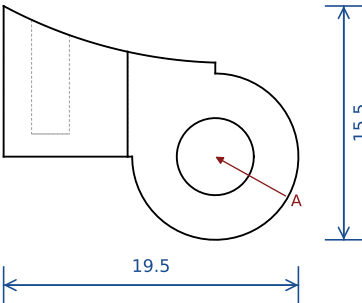
TOP



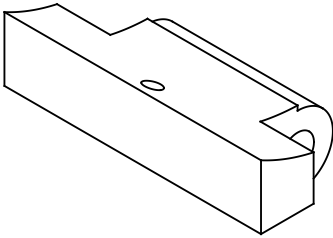
FRONT



RIGHT



ISO (not to scale)



NOTES

- Body rides C2's OD (saddle R31.8); lug is round about the pin
- Body bottom = pin height by design (swing clearance) - do not extend it
- Lug spans x 66.3..89.7 (0.3 end float between A3's lugs)

HOLE / FEATURE SCHEDULE

A: Ø5.1 pin bore THRU the lug, axis at (y -1, z -38) @ along x

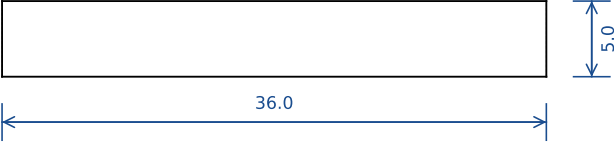
B: 2x M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG), from the saddle side @ x 78, 102 at y=-11.9

GENERAL

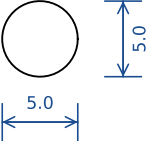
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.  
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.  
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

|                                                                                            |                                    |                                                                                           |                                              |
|--------------------------------------------------------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>RFID BIKE LOCK - CNC CASING</b><br>cad/cnc_casing_cq.py rev 3b                          | MATERIAL<br>6061-T6 (PETG stage 1) | SCALE (ortho views)<br>2.00 : 1                                                           | DATE<br>2026-09-07                           |
| <b>H - hinge block (on C2, carries C2's lug)</b><br>file: cnc-design/step/hinge_block.step | QTY<br>1                           | BOUNDING BOX x / y / z<br>40.0 / 19.5 / 15.5 (x 58.0..98.0, y -15.0..4.5, z -43.5..-28.0) | SHEET<br>4 / 12<br>3rd angle<br>A3 landscape |

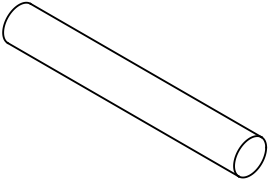
FRONT



RIGHT



ISO (not to scale)



NOTES

- Standard Ø5 x 36 dowel (cut a 40 to length if needed); ends chamfered
- Assembly: enter through A3 lug 1 (x 58 face) to the blind end, then press a Ø5 x 2 plug flush behind it

HOLE / FEATURE SCHEDULE

A: Ø5 h6 x 36

GENERAL

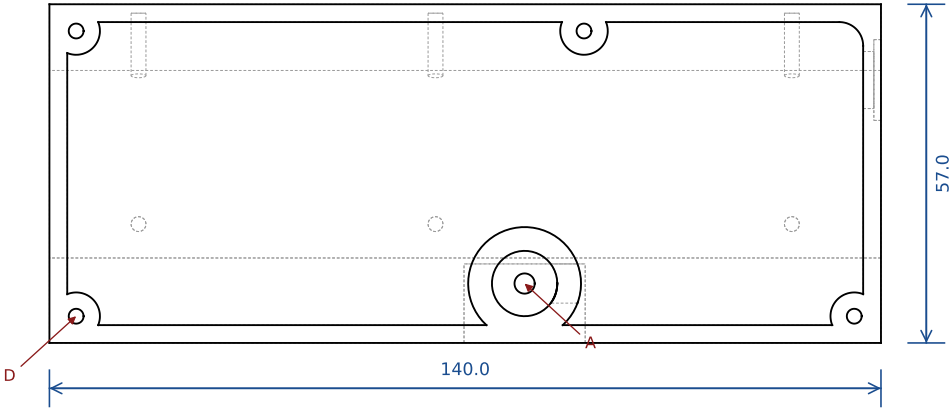
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.

Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.

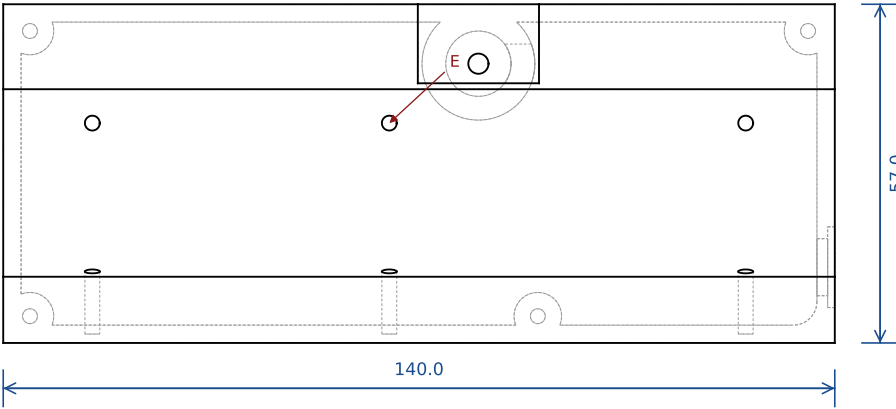
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

|                                                                               |                                                  |                                                                                                           |                                                                                 |
|-------------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <div>RFID BIKE LOCK - CNC CASING</div> <div>cad/cnc_casing_cq.py rev 3b</div> | <div>MATERIAL</div> <div>steel dowel Ø5 h6</div> | <div>SCALE (ortho views)</div> <div>2.00 : 1</div>                                                        | <div>DATE</div> <div>2026-09-07</div>                                           |
| <div>hinge pin</div> <div>file: cnc-design/step/hinge_pin.step</div>          | <div>QTY</div> <div>1</div>                      | <div>BOUNDING BOX x / y / z</div> <div>36.0 / 5.0 / 5.0 (x 60.0..96.0, y -3.5..1.5, z -40.5..-35.5)</div> | <div>SHEET</div> <div>5 / 12</div> <div>3rd angle</div> <div>A3 landscape</div> |

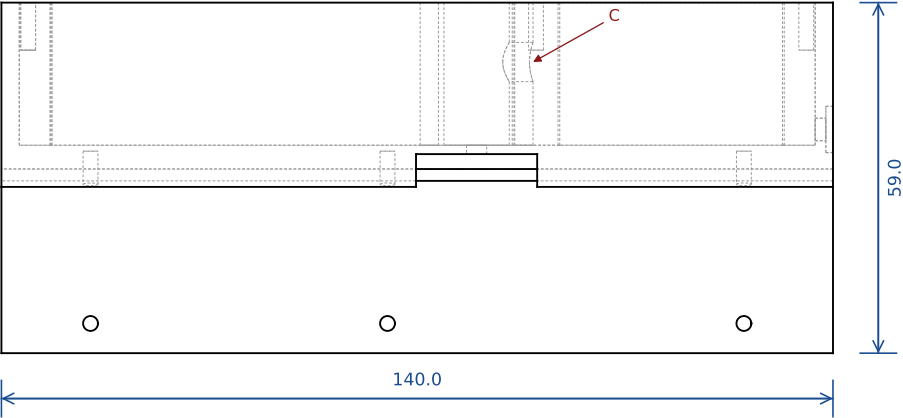
TOP



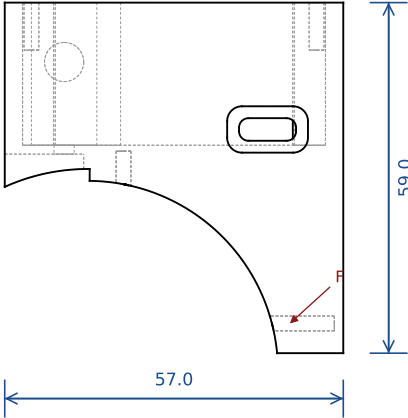
BOTTOM



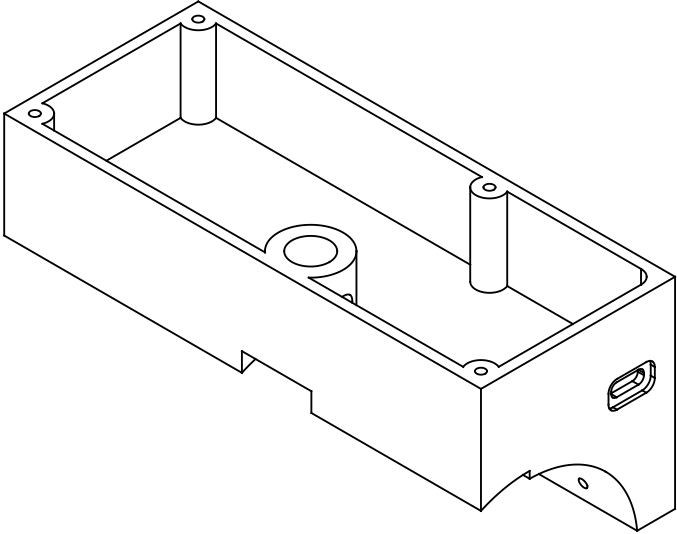
FRONT



RIGHT



ISO (not to scale)



NOTES

- Underside: saddle R32.0 on the C1 side, relieved to R34.0 over C2 (y<0.3) - C2's rim arcs outward as it swings
- Interior pocket 134 x 51 x 24 deep from z=38, R6 corners
- Closure-block pocket under the -y overhang: x 79.8..100.2, open toward -y, roof at z=36.5
- Latch boss Ø19 rises from the floor around the receiver bore

HOLE / FEATURE SCHEDULE

- A: Ø11 receiver bore from the top down to the floor (z 38) @ (90, -4)
- B: Ø3.4 closure-screw clearance THRU the floor (1.5 thick) under A @ same axis
- C: Ø6.6 plunger channel, horizontal along +x from the bore, axis z=52 @ (x 90.., y -4)
- D: 4x M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG) lid screws, 8 deep from the top @ (14.5, -9.5), (14.5, 38.5), (145.5, -9.5), (100, 38.5)
- E: 3x M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG) vertical from the saddle at y=+6 (crown row), to z=37 @ x 25, 75, 135
- F: 3x M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG) horizontal (+y) at z=+8 into the skirt (skirt row) @ x 25, 75, 135
- G: USB-C slot 9.6 x 3.8 THRU the +x end wall, R1.6, + plug recess 13.6 x 7.8 x 2 deep outside @ y 30.25, z 40.65

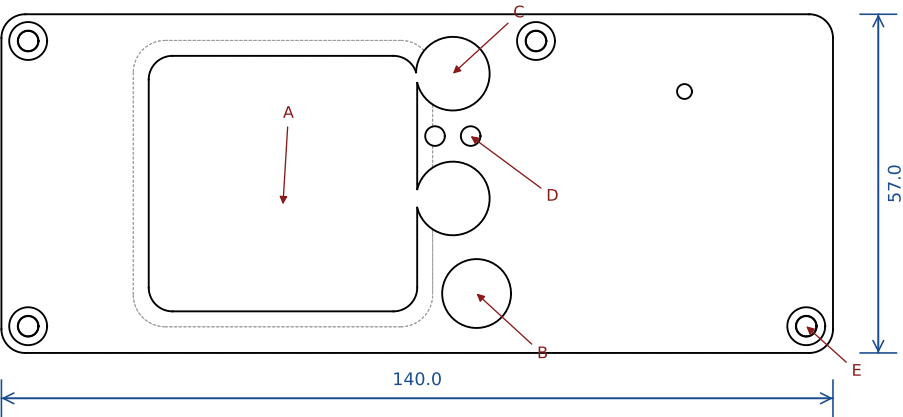
GENERAL

Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.  
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.  
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

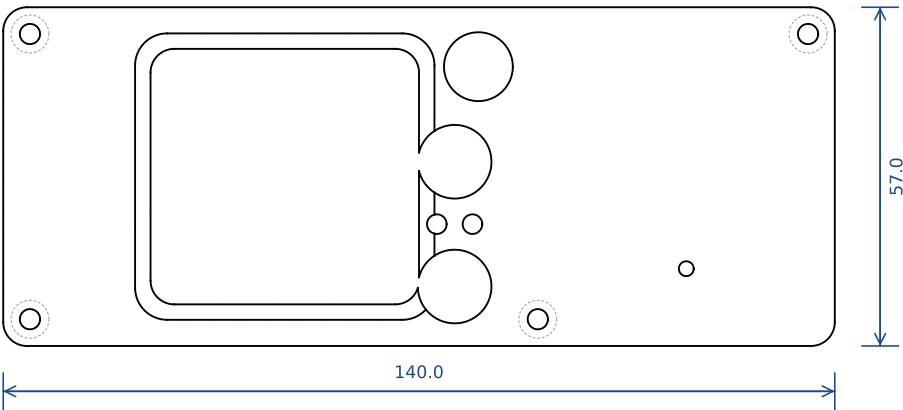
|                                                                                    |                                           |                                                                                           |                                              |
|------------------------------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>RFID BIKE LOCK - CNC CASING</b><br>cad/cnc_casing_cq.py rev 3b                  | MATERIAL<br>6061-T6 billet (PETG stage 1) | SCALE (ortho views)<br>0.79 : 1                                                           | DATE<br>2026-09-07                           |
| <b>A1 - top box (latch + electronics)</b><br>file: cnc-design/step/A1_top_box.step | QTY<br>1                                  | BOUNDING BOX x / y / z<br>140.0 / 57.0 / 59.0 (x 10.0..150.0, y -14.0..43.0, z 3.0..62.0) | SHEET<br>6 / 12<br>3rd angle<br>A3 landscape |



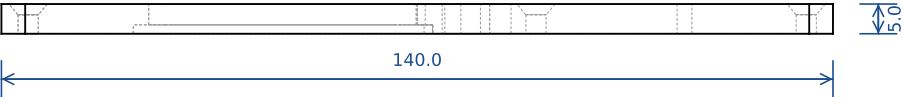
TOP



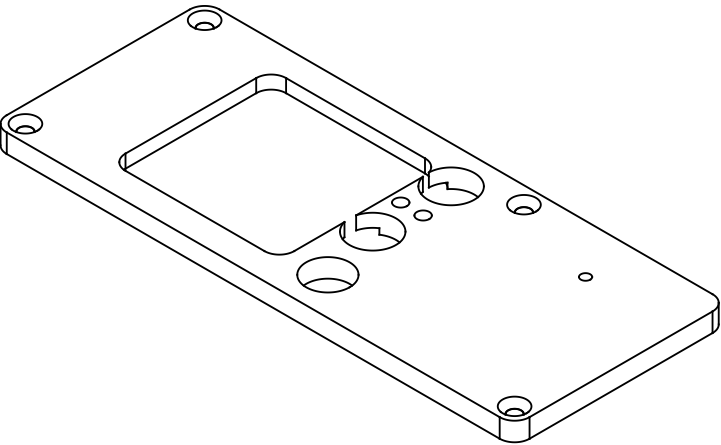
BOTTOM



FRONT



ISO (not to scale)



NOTES

- RF window 45.2 x 43 THRU, R4 corners; underside recess 50.4 x 48.2 x 1.5 deep for the insert flange
- Lid screws countersunk 90deg from the top for M3 flat head

HOLE / FEATURE SCHEDULE

- A: window 45.2 x 43 THRU @ centre (57.4, 14.5)
- B: Ø11.6 THRU (latch bore pass-through) @ (90, -4)
- C: 2x Ø12.4 THRU (sealed buttons: green = wake, red = cancel/admin) @ (86, 33) (86, 12)
- D: 2x Ø3.3 THRU (LEDs) + Ø2.5 buzzer sound hole @ (125, 30) @ (83, 22.5), (89, 22.5)
- E: 4x Ø3.4 THRU, 90deg C'SINK Ø6.4 from the top @ (14.5, -9.5), (14.5, 38.5), (145.5, -9.5), (100, 38.5)

GENERAL

Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.  
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.  
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

RFID BIKE LOCK - CNC CASING

cad/cnc\_casing\_cq.py rev 3b

MATERIAL  
6061 plate 5 mm (PETG stage 1)

SCALE (ortho views)  
0.79 : 1

DATE  
2026-09-07

A2 - lid plate

file: cnc-design/step/A2\_lid.step

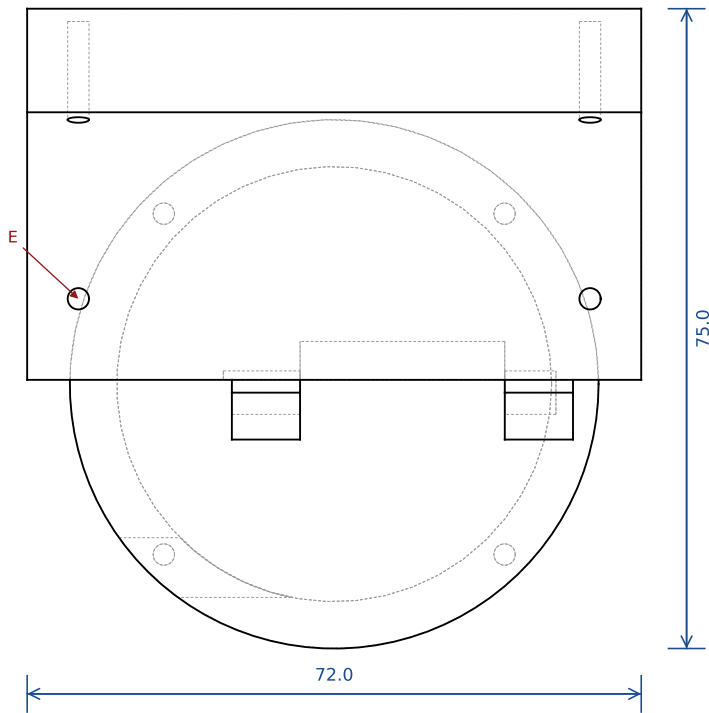
QTY  
1

BOUNDING BOX x / y / z  
140.0 / 57.0 / 5.0 (x 10.0..150.0, y -14.0..43.0, z 62.0..67.0)

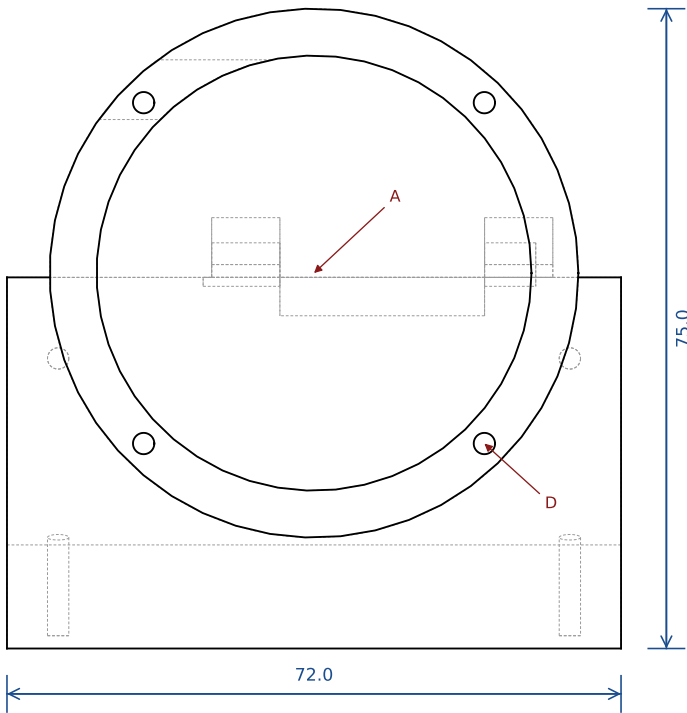
SHEET  
7 / 12

3rd angle  
A3 landscape

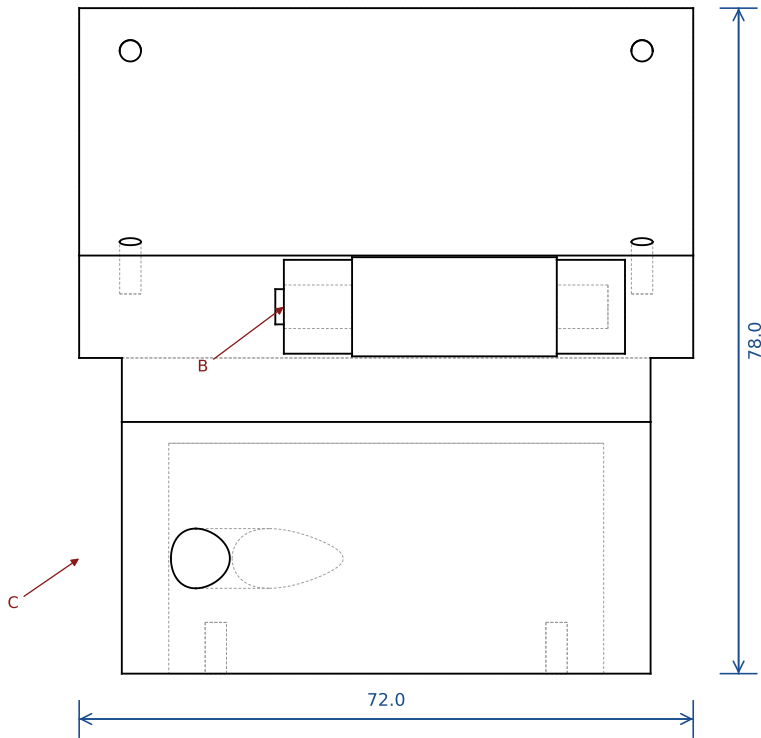
TOP



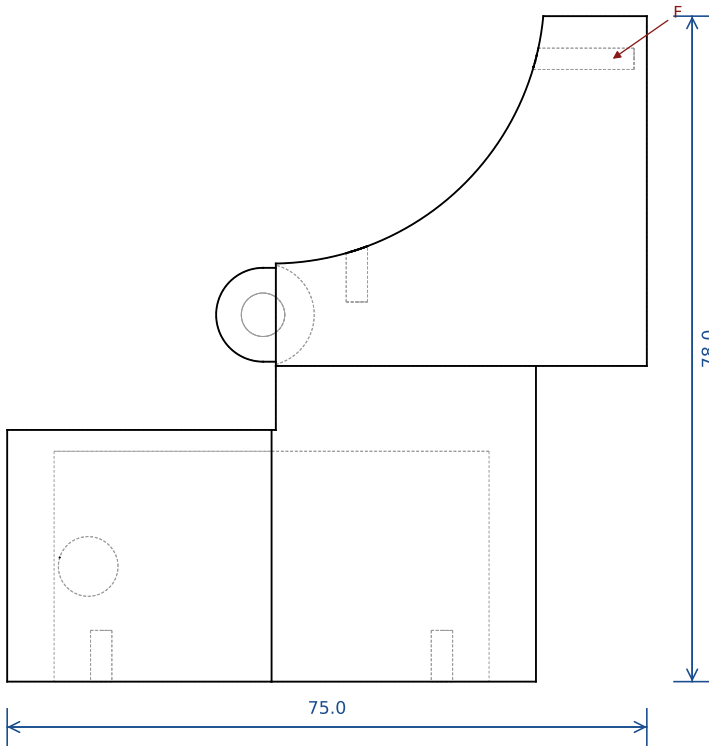
BOTTOM



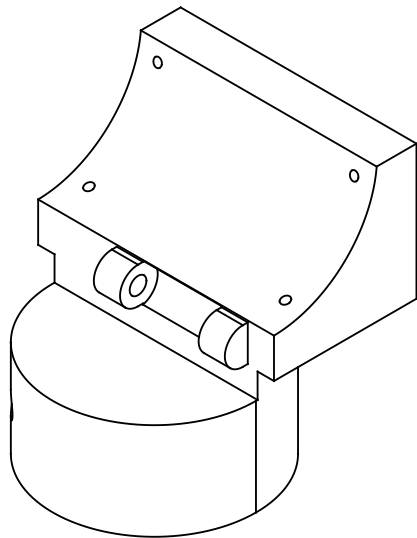
FRONT



RIGHT



ISO (not to scale)



NOTES

- Cradle x 34..106, y 0.5..44, saddle R32.0 on top; skirt rib y 32..44 up to z=-3
- Puck Ø62 centred under the tube (x 70, y 0); flat top at z=-51.5 on the C2 side = hinge STOP
- Two hinge lugs Ø11 about the pin axis (y -1, z -38) at x 58..66 and 90..98; Ø12 clearance between them for C2's lug
- Spool cartridge + power spring load from below; A4 closes the pocket

HOLE / FEATURE SCHEDULE

A: Ø51 spool pocket from the bottom face up to z=-54 (depth 27) @ axis (70, 0)

B: Ø5.1 pin bore along x from lug 1's outer face (x 58), BLIND, ends at x 96 @ axis (y -1, z -38)

C: Ø7 cable exit along x through the puck wall (fit a steel bushing) @ y -21.5, z -67.5

D: 4x M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG) cover screws, 6 deep from the bottom face, in the puck wall @ PCD 56.5 at 45/135/225/315deg

E: 2x M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG) vertical from the saddle at y=10 (crown row) @ x 40, 100

F: 2x M3 tapped (Ø2.5 pilot modeled; tap M3x0.5 in metal, self-tap in PETG) horizontal (+y) at z=-8 into the skirt rib (skirt row) @ x 40, 100

GENERAL

Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.  
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.  
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

RFID BIKE LOCK - CNC CASING

cad/cnc\_casing\_cq.py rev 3b

MATERIAL

6061-T6 billet (PETG stage 1)

SCALE (ortho views)

1.13 : 1

DATE

2026-09-07

A3 - spool puck + cradle (hinge knuckles)

file: cnc-design/step/A3\_bottom\_box.step

QTY

1

BOUNDING BOX x / y / z

72.0 / 75.0 / 78.0 (x 34.0..106.0, y -31.0..44.0, z -81.0..-3.0)

SHEET

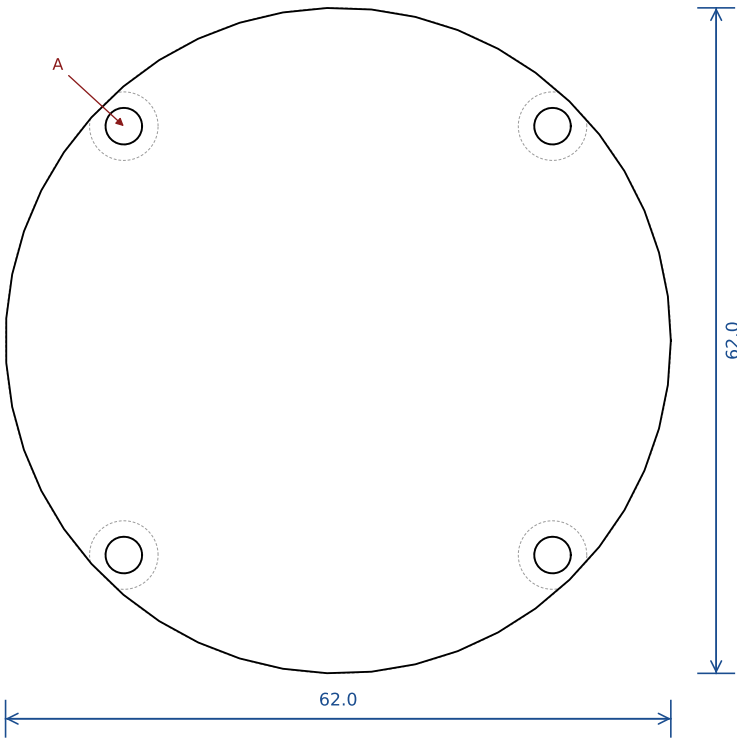
8 / 12

3rd angle

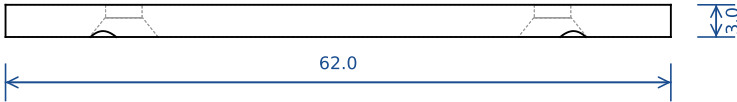
A3 landscape



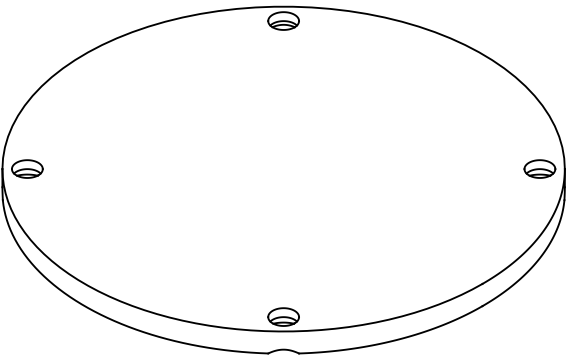
TOP



FRONT



ISO (not to scale)



NOTES

- Ø62 disc, 3 thick; gasket under it
- Screws countersunk 90deg from the outside for M3 flat head

HOLE / FEATURE SCHEDULE

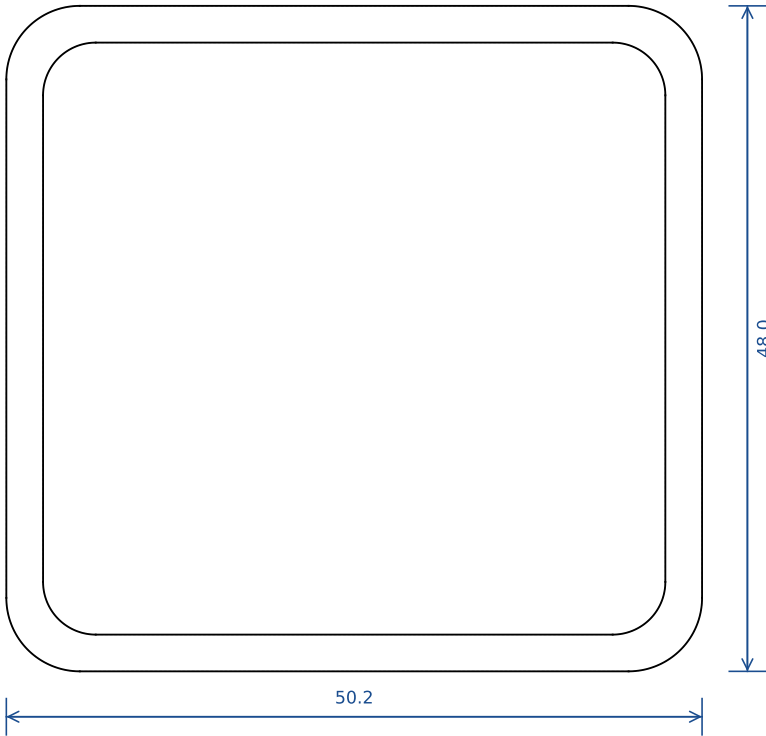
A: 4x Ø3.4 THRU, 90deg C'SINK Ø6.4 @ PCD 56.5 at 45/135/225/315deg

GENERAL

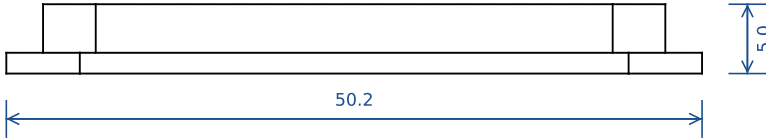
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.  
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.  
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

|                                                                     |                                            |                                                                                            |                                              |
|---------------------------------------------------------------------|--------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>RFID BIKE LOCK - CNC CASING</b><br>cad/cnc_casing_cq.py rev 3b   | MATERIAL<br>6061 plate 3 mm (PETG stage 1) | SCALE (ortho views)<br>1.42 : 1                                                            | DATE<br>2026-09-07                           |
| <b>A4 - puck cover</b><br>file: cnc-design/step/A4_cover_plate.step | QTY<br>1                                   | BOUNDING BOX x / y / z<br>62.0 / 62.0 / 3.0 (x 39.0..101.0, y -31.0..31.0, z -84.0..-81.0) | SHEET<br>9 / 12<br>3rd angle<br>A3 landscape |

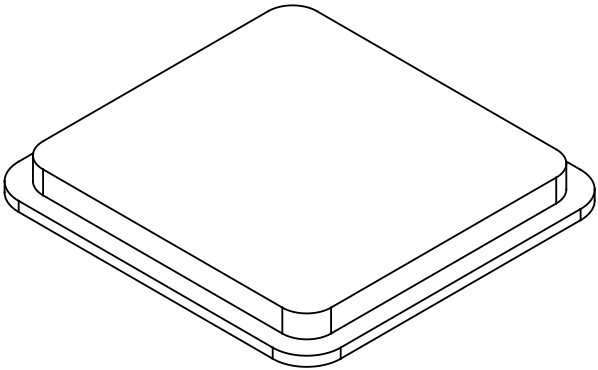
TOP



FRONT



ISO (not to scale)



NOTES

- Body fills the lid cutout flush with the lid top; flange sits in the lid's underside recess
- Opaque by choice - RF does not care; keep it non-metallic

HOLE / FEATURE SCHEDULE

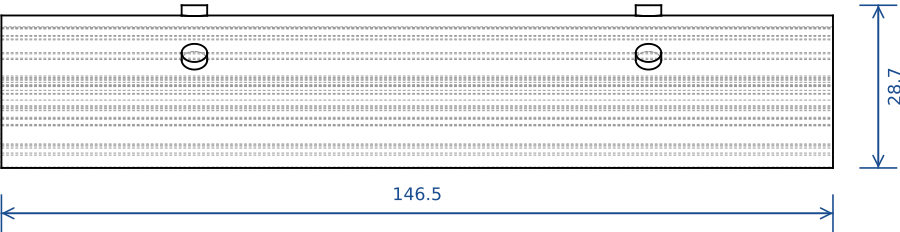
A: body 44.9 x 42.7 x 3.5, flange 50.2 x 48 x 1.5

GENERAL

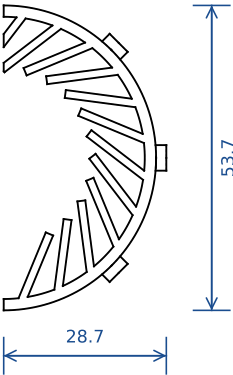
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.  
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.  
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

|                                                                                                  |                                                                  |                                                                                                           |                                                                                  |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <div>RFID BIKE LOCK - CNC CASING</div> <div>cad/cnc_casing_cq.py rev 3b</div>                    | <div>MATERIAL</div> <div>printed PETG/ASA or polycarbonate</div> | <div>SCALE (ortho views)</div> <div>1.83 : 1</div>                                                        | <div>DATE</div> <div>2026-09-07</div>                                            |
| <div>A5 - RF window insert (opaque)</div> <div>file: cnc-design/step/A5_window_insert.step</div> | <div>QTY</div> <div>1</div>                                      | <div>BOUNDING BOX x / y / z</div> <div>50.2 / 48.0 / 5.0 (x 32.3..82.5, y -9.5..38.5, z 62.0..67.0)</div> | <div>SHEET</div> <div>10 / 12</div> <div>3rd angle</div> <div>A3 landscape</div> |

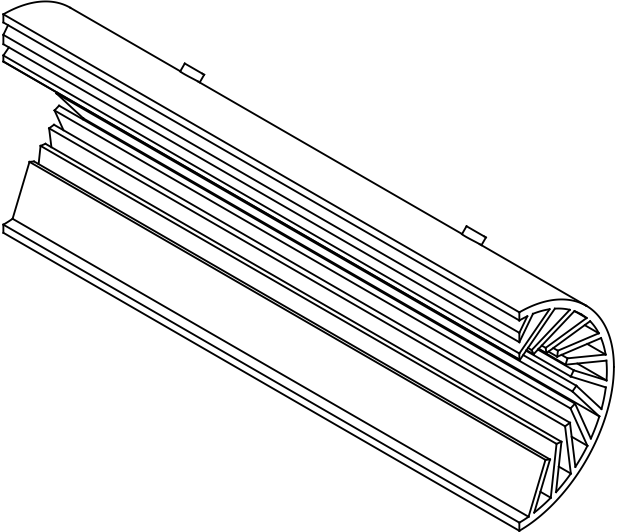
TOP



RIGHT



ISO (not to scale)



NOTES

- Base ring R24.85..26.85 with 24 fins (1.2 wide x 4 tall) - fins bend to grip Ø32..46 down tubes
- Left half is the mirror (liner\_left)

HOLE / FEATURE SCHEDULE

A: 24 fins on 15deg pitch, 1.4 x 12, leaning 30deg

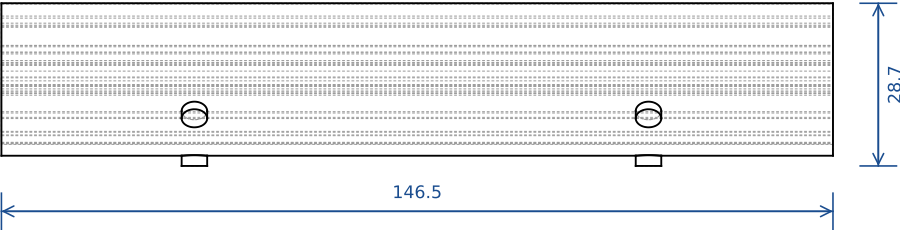
B: 6x press-fit studs Ø4.5 ±0.1 x 1.8 on the outside; mate: Ø4.2 +0.1/-0 x 2 blind holes in the tube wall (0.3 interference) @ x 35, 115 at -45/0/+45 deg from the mid-arc (6 per half)

GENERAL

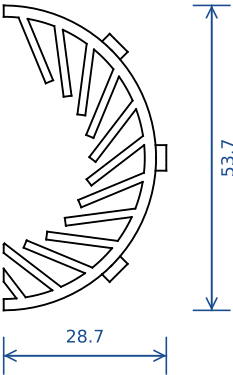
Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.  
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.  
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

|                                                                              |                                               |                                                                                           |                                               |
|------------------------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>RFID BIKE LOCK - CNC CASING</b><br>cad/cnc_casing_cq.py rev 3b            | MATERIAL<br>TPU 95A, printed on its seam face | SCALE (ortho views)<br>0.75 : 1                                                           | DATE<br>2026-09-07                            |
| <b>liner (right half, C1 side)</b><br>file: cnc-design/step/liner_right.step | QTY<br>1                                      | BOUNDING BOX x / y / z<br>146.5 / 28.7 / 53.7 (x 1.0..147.5, y -0.0..28.7, z -26.9..26.9) | SHEET<br>11 / 12<br>3rd angle<br>A3 landscape |

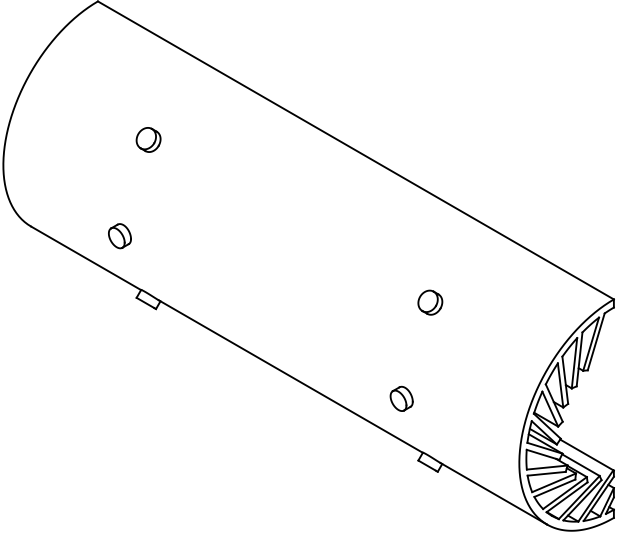
TOP



LEFT



ISO (not to scale)



NOTES

- Mirror of liner\_right

HOLE / FEATURE SCHEDULE

A: 24 fins on 15deg pitch, 1.4 x 12, leaning 30deg

B: 6x press-fit studs Ø4.5 x 1.8 - see liner\_right

GENERAL

Units mm. Datum: tube axis = X, seam plane y=0, Z up. Break sharp edges 0.3.  
Stage 1 (PETG): print on the face noted, pilots self-tap M3. Metal: tap M3x0.5.  
Internal corners R4 unless the model shows R6. Tolerance ±0.1 on fits, ±0.3 elsewhere.

|                                                                            |                                               |                                                                                           |                                               |
|----------------------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>RFID BIKE LOCK - CNC CASING</b><br>cad/cnc_casing_cq.py rev 3b          | MATERIAL<br>TPU 95A, printed on its seam face | SCALE (ortho views)<br>0.75 : 1                                                           | DATE<br>2026-09-07                            |
| <b>liner (left half, C2 side)</b><br>file: cnc-design/step/liner_left.step | QTY<br>1                                      | BOUNDING BOX x / y / z<br>146.5 / 28.7 / 53.7 (x 1.0..147.5, y -28.7..0.0, z -26.9..26.9) | SHEET<br>12 / 12<br>3rd angle<br>A3 landscape |